```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import time
```

```
df=pd.read_csv(r"/content/diabetes.csv")
```

```
df.columns
```

```
Index(['pr', 'glucose', 'bp_diastolic', 'skin_triceps', 'insulin', 'bmi',
       'pedigree', 'age', 'label'],
      dtype='object')
```

```
df.rename(columns={'label':'is_diabetic'},inplace=True)
```

```
df.columns
```

```
Index(['pr', 'glucose', 'bp_diastolic', 'skin_triceps', 'insulin', 'bmi',
       'pedigree', 'age', 'is_diabetic'],
      dtype='object')
```

```
X=df.drop('glucose',axis=1)
y=df['glucose']
```

```
X.columns
```

```
Index(['pr', 'bp_diastolic', 'skin_triceps', 'insulin', 'bmi', 'pedigree',
       'age', 'is_diabetic'],
      dtype='object')
```

```
print(df.shape)
df.head()
```

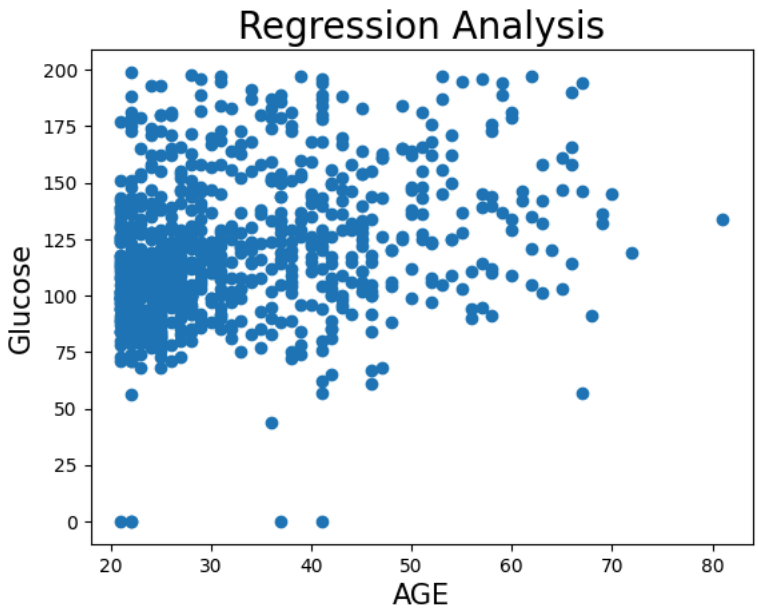
```
(768, 9)
```

	pr	glucose	bp_diastolic	skin_triceps	insulin	bmi	pedigree	age	is_diabetic
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Next steps: Generate code with df New interactive sheet

```
y1 = df['glucose']     # target
x1 = df['age']         # feature
```

```
plt.scatter(x1, y1)
plt.title("Regression Analysis", fontsize=20)
plt.xlabel("AGE", fontsize=15)
plt.ylabel("Glucose", fontsize=15)
plt.show()
```




```
from scipy.stats import linregress

slope, intercept, r_value, p_value, std_err = linregress(x1, y1)

y2 = slope * x1 + intercept
```

```
fig = plt.figure()

plt.scatter(x1, y1)
plt.plot(x1, y2, lw=4, c='orange')

plt.title("Regression Analysis", fontsize=20)
plt.xlabel("AGE", fontsize=15)
plt.ylabel("Glucose", fontsize=15)

plt.show()
```

```
givenAge = 40
predGlucose = slope * givenAge + intercept
print(predGlucose)
```

```
125.73688327791153
```

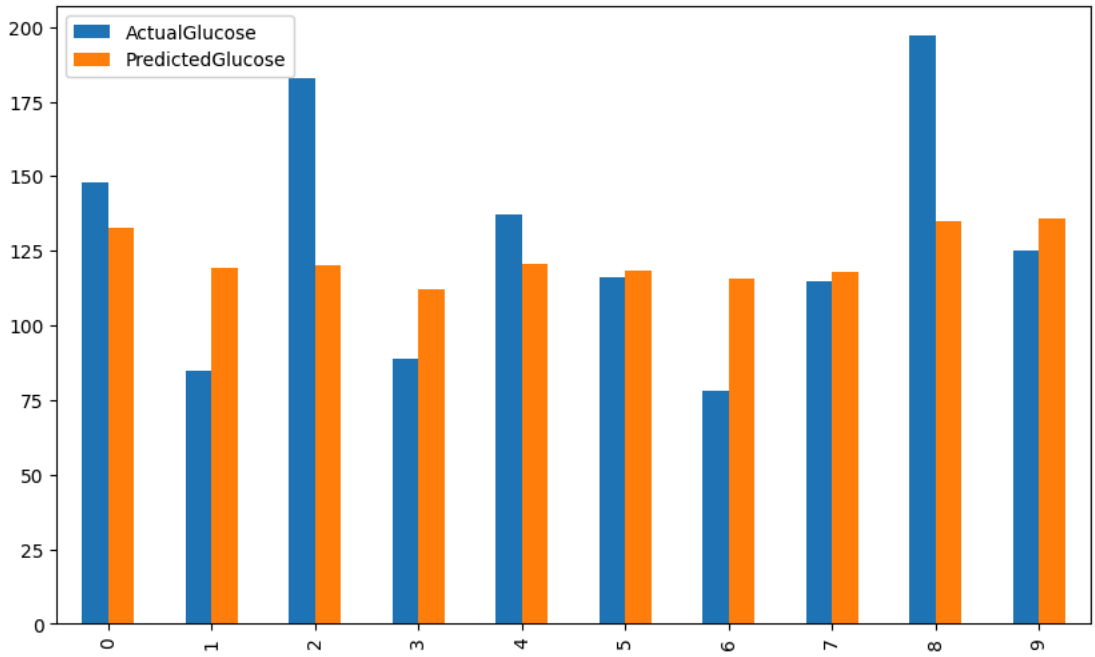
```
df_p = pd.DataFrame({"ActualGlucose": y1, "PredictedGlucose": y2})
df_p.head()
```

	ActualGlucose	PredictedGlucose
0	148	132.901064
1	85	119.289120
2	183	120.005538
3	89	112.124939
4	137	120.721957

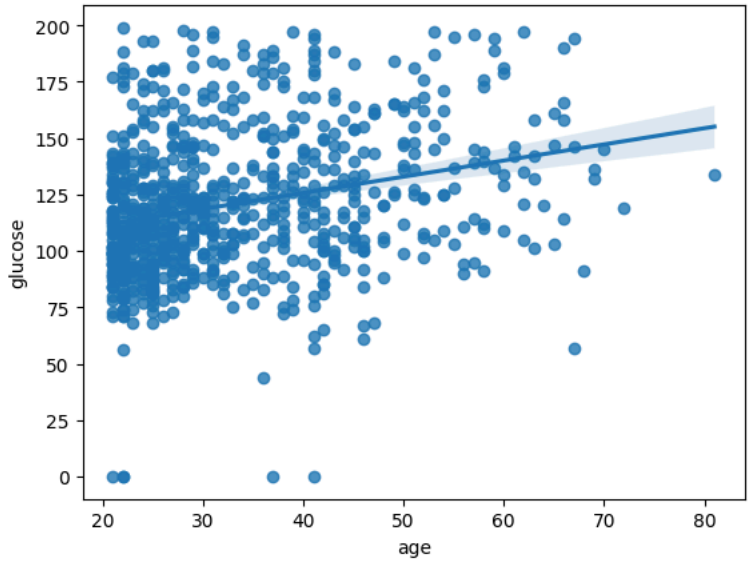
Next steps: Generate code with df_p New interactive sheet

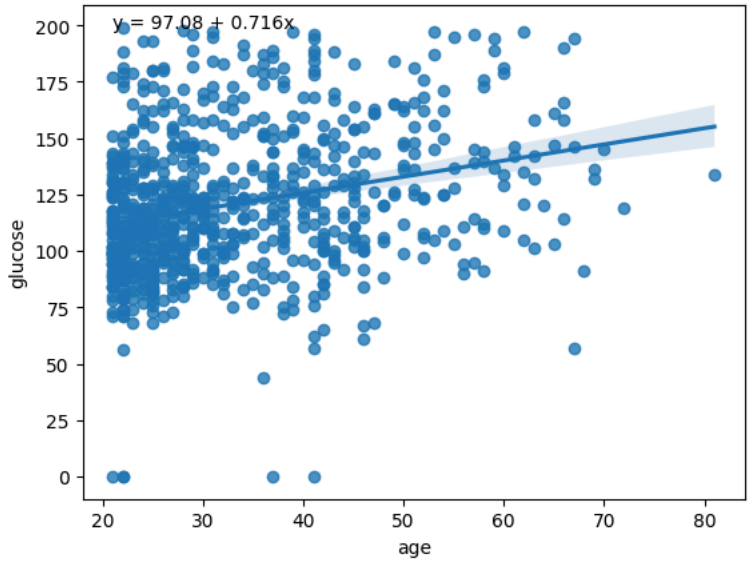
```
df_p.head(10).plot(kind="bar", figsize=(10, 6))
```

```
<Axes: >
```




```
import seaborn as sns
sns.regplot(data=df, x=df.age, y=df.glucose)
plt.show()
```




```
plt.text(min(df.age), max(df.glucose),
```

```
'y = ' + str(round(intercept,3)) + ' + ' + str(round(slope,3)) + 'x')
sns.regplot(data=df, x=df.age, y=df.glucose)
plt.show()
```